

## 1 Data Preparation

- TriviaQA rc.nocontext
- 2,000 train / 200 eval questions
- K<sub>0</sub> factual baseline (held-out)
- Bidirectional substring matching

## 2 Recursive Training

- QLoRA 4-bit NF4 / FFT
- Fresh adapter each generation
- 10 generations  $G_1 \rightarrow G_{10}$
- 3 backbones: Qwen, Gemma 3, 4

*↻ Output Gen  $t \rightarrow$  training for Gen  $t+1$*

## 3 Pressure Axes

- Adapter rank — 4, 16, 32, 64, 128, 256
- Learning rate — 1e-6 to 2e-5
- Rank × LR joint-dependence grid
- Exposure — 0/10/25/50% removal

## 4 Multi-Metric Evaluation

- K<sub>0</sub> retention — factual preservation
- Distribution shift — Distinct-n
- Lexical diversity — MTLD, Stopword Ratio
- Effective rank —  $\exp(H(\sigma))$  of B×A

## 5 Regime Classification

**Homeostatic** — retention > 90%, stable

**Bounded**  
— stable retention, distribution drifts

**Degradative**  
— retention declines ~1–3 pp/gen

## 6 Pressure-Regime Mapping

- Effective Training Pressure (ETP) threshold
- Regime transitions: threshold-like within grid
- Cross-backbone generalization
- Rank, LR, exposure → unified pressure